

Airframe Technology Master

S07-L05

Interface Control Document

Álvaro Ortiz del Cerro
May 20th 2025

Content

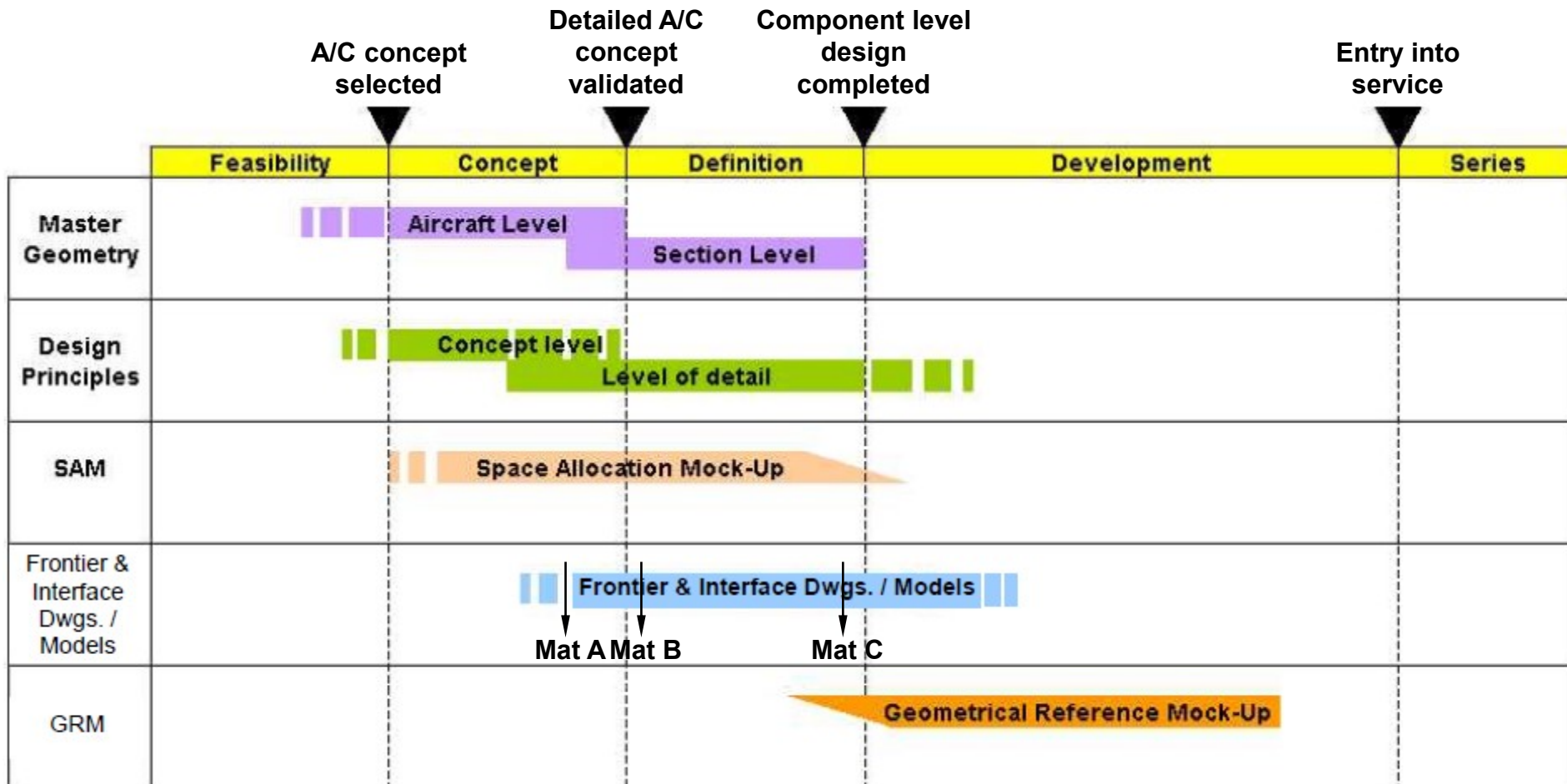
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Purpose and scope

- For cost and time reasons, Aircraft Work-packages have to be manufactured in parallel.
- Aircraft work sharing imply several countries and several manufacturing plants all over the world, documents shall be created to ensure the Aircraft quality and the Customer satisfaction.
- Interface documents exists to ensure at each level of assembly the quality of the whole final Aircraft.

Purpose and scope

- The interface definition process is launched during the concept phase when master geometry and design principles have a minimum maturity level.



Interface contents

- Description and geometrical features:
 - Views of the interface assembly (all parts) including cross section showing the two mated parts.
 - Vectors visual definition to ensure accurate understanding of the interface orientation.
 - Stack-up requirement identification (geometrical, displacements, thermal...)
 - Size and geometrical tolerances (relative to dedicated datum system) allocated the involved interface items, including stack-up tolerances and the associated responsibility sharing.
 - Functional clearances between both sides of the interface. (Hinges gaps, bumpers gaps, mounts links gaps... etc)
 - Aerosmoothness steps and gaps ranges between all components and singularities (deviations from lofted contour)
 - Theoretical nominal geometric definition of the interface (mounting, connecting...)
 - DMU model number or drawing used for the generation of the data.
- Mechanical properties:
 - Seals or springs definition and functional crushing ranges. (Nominal, min and max)
 - Threaded items installation coupling torque range (min and max) if necessary to the definition interface and minimum engagement for proper locking feature operation and maximum engagement to prevent clash (envelope definition).
- Materials:
 - Generic designations of materials, protections, final paint and contour protections thicknesses.
 - Lubricant designations, if necessary, to the definition of interface.
 - Standard items references (bolts, electrical and fluids connectors, etc)

KEY CHARACTERISTICS AND TOLERANCES

Interface contents

— Tolerances

- Tolerances are the defined allowable deviation from nominal of parts or assemblies geometrical dimensions.
- Tolerances are required because of the inability of any process to manufacture parts or assemblies to their nominal geometry.
- The tolerances shall define the maximum permissible limits for which the geometric functional requirements are still ensured.
- Tolerances also includes measurement systems, which cannot inspect parts to nominal dimensions because of the variability in the measurement equipment.
- The tolerances described in the drawings shall be ensured and measured in the same configuration as during the integration of the component at the next assembly level above.

Definition

- Prior to any data being embodied on an interface document, all functions withing the work package shall review and agree the Geometric Key Characteristics & Tolerancing information.
- 3 kind of features must be identified taken into account the criticality:
 - **Key characteristics (KCs):** are all those requirements that validate Installation and Performance.
 - Are representing a quantifiable geometrical feature for which any deviation from nominal has a significant effect on fit, form and function of the product.
 - All KCs shall be included in the Interface Control documents.
 - **Critical Item (CTI):** is an important geometrical feature for which results are within the specification (tolerance) defined regardless of where these results lay.
 - **Specific Tolerance Requirement (STR):** is a geometrical feature which does not have a significant impact on a product's performance but requires specific attention throughout the development of the product
 - STR is to be controlled for specific reason (rigging, hole-to-edge distance...) and shall be recorded in the Product Cascade.

TYPES

— KCs

- Performance Key Characteristics (PKC)
 - Those linked to Installation and Performance Requirements
 - They are achieved at Installation Level
 - The Tolerance Chain can be broken down to subordinated contributing Key Characteristics (AKCs MKCs)
- Assembly Key Characteristics (AKC)
 - They are achieved at Component Assembly Level
 - They can be broken down to subordinated contributing key characteristics (other AKCs and/or MKCs)
- Manufacturing Key Characteristics (MKC)
 - They are characteristics of each part, achieved during ManufacturingKey characteristics (KCs)

TYPES

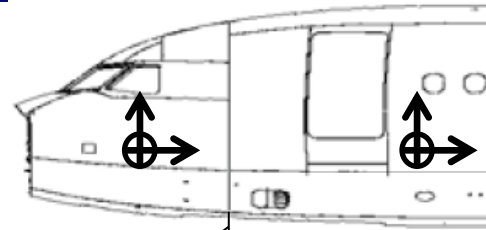
— CTIs

- Performance Critical Item (PCTI)
 - It is a top level geometrical Critical Item that governs the overall design intent and aircraft performance
- Assembly Critical Item (ACTI)
 - It is the next level of geometrical feature (below PCTI) that covers all levels of Assembly
- Manufacturing Critical Item (MCTI)
 - It is a physical geometrical feature at Part level that needs to be achieved in order to realize Performance and Assembly Critical Items

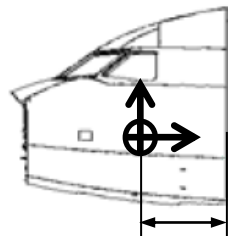
— STRs

- Assembly Tolerance Requirement (ATR)
 - It is the next level of geometrical feature (below STR) that covers all levels of Assembly.
- Manufacturing Tolerance Requirement (MTR)
 - It is a physical geometrical feature at Part level that needs to be achieved in order to realize Specific and Assembly Tolerance Requirement

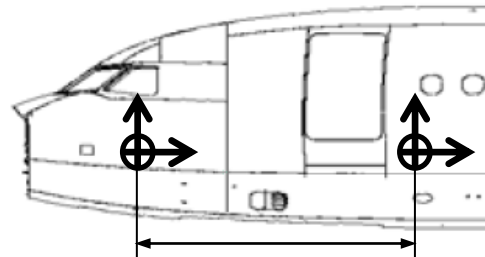
Interface Control Document



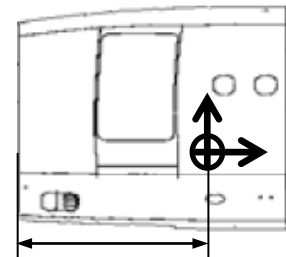
PKC $A \pm 2$



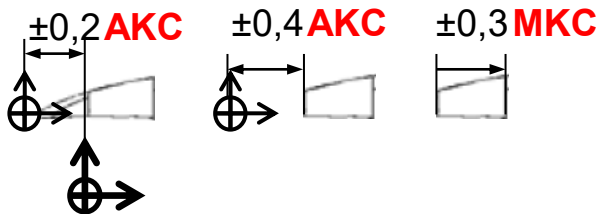
AKC $H \pm 0,9$



AKC $\pm 0,2$



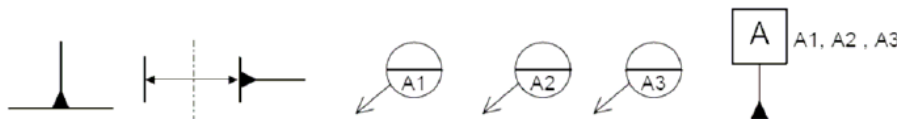
AKC $J \pm 0,9$



Interface contents

— Datums

- Datum are defined on every package in order to guide their assembly processes.
- Every requirement/tolerance shall be expressed in a defined datum.
- Datum can be physical as lines, planes, holes... or non physical as mean planes or lines between, laser targets...
- The representation of the datum is made following the ISO1101 rules.



- The datum system shall stop the 6 freedoms (3 rotations and 3 translations).
→ The datum system shall refer to the physical or measured features that stop those freedoms (plane, line, axis, point, specific coordinates, etc.).
- The ICD datum gives the starting point to make a measurement and/or an assembly, and are different from the ones defined in the Master Geometry which are only a co-ordinate system.

WINGLET AXIS SYSTEM REF DATUM

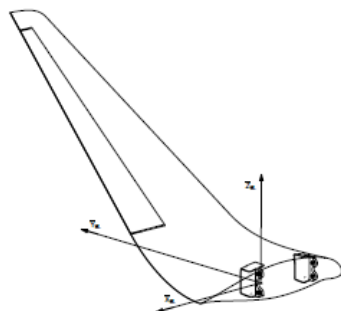
	TX _{WL}	TY _{WL}	TZ _{WL}	RX _{WL}	RY _{WL}	RZ _{WL}
◇						
/						
•						

- ◇ = MEAN PLANE DEFINED BY 8 POINTS: P1_FWD, P1_AFT THROUGH P4_FWD, P4_AFT
- / = PROJECTION OF :
 - BISECTIG LINE DEFINED BY SEGMENT [P3_FWD, P3_AFT] AND SEGMENT [P4_FWD, P4_AFT] ON
 - PLANE PREVIOUSLY DEFINED
- = INTERSECTION OF:
 - LINE DEFINED BY [PROJECTION OF P3_FWD ON PLANE] AND [PROJECTION OF P4_FWD ON PLANE] AND
 - LINE PREVIOUSLY DEFINED

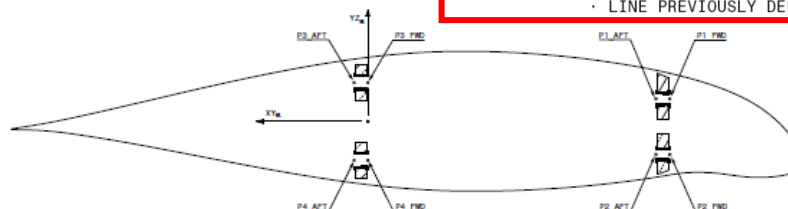
	TX _{WL}	TY _{WL}	TZ _{WL}	RX _{WL}	RY _{WL}	RZ _{WL}
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/						
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 - PLANE PREVIOUSLY DEFINED
- = INTERSECTION OF:
 - LINE DEFINED BY [PROJECTION OF P3_FWD ON PLANE] AND [PROJECTION OF P4_FWD ON PLANE] AND
 - LINE PREVIOUSLY DEFINED

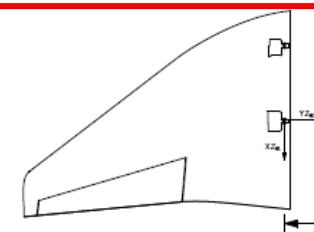
INTERFACE POINT	X _{INT}	Y _{INT}	Z _{INT}	MG DESIGNATION
P1_FWD	-453.003	0	0	100
P1_AFT	-453.003	0	0	100
P2_FWD	0	0	0	100
P2_AFT	23.2	0	0	100
P3_FWD	0	0	0	100
P3_AFT	23.2	0	0	100
P4_FWD	0	0	0	100
P4_AFT	23.2	0	0	100



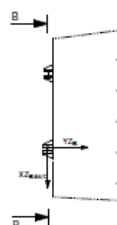
WINGLET ISO VIEW
NTS



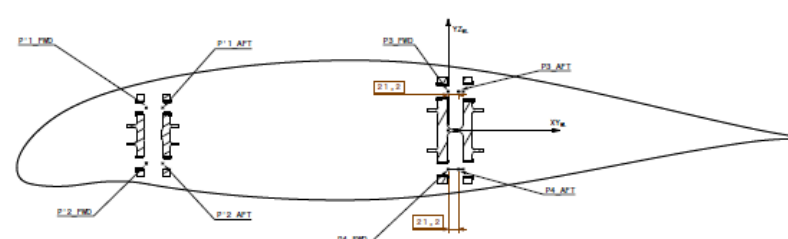
SECTION A-A
SCALE 2:5



WINGLET TOP VIEW



WINGLET COUNTERPART AT A/C
TOP VIEW



SECTION B-B
SCALE 2:5



WINGLET COUNTERPART ISO VIEW
NTS

	TX _{WL}	TY _{WL}	TZ _{WL}	RX _{WL}	RY _{WL}	RZ _{WL}
◇						
/						
•						

- ◇ = MEAN PLANE DEFINED BY 8 POINTS: P1_FWD, P1_AFT THROUGH P4_FWD, P4_AFT
- / = PROJECTION OF :
 - BISECTIG LINE DEFINED BY SEGMENT [P3_FWD, P3_AFT] AND SEGMENT [P4_FWD, P4_AFT] ON
 - PLANE PREVIOUSLY DEFINED
- = INTERSECTION OF:
 - LINE DEFINED BY [PROJECTION OF P3_FWD ON PLANE] AND [PROJECTION OF P4_FWD ON PLANE] AND
 - LINE PREVIOUSLY DEFINED

INTERFACE POINT	X _{INT}	Y _{INT}	Z _{INT}	MG DESIGNATION
P1_FWD	0	34.407	100	100
P1_AFT	0	34.407	100	100
P2_FWD	0	-50.619	100	100
P2_AFT	0	-50.619	100	100
P3_FWD	0	50.600	100	100
P3_AFT	23.2	0	50.600	100
P4_FWD	0	50.600	100	100
P4_AFT	23.2	0	-50.600	100

COMPUTER PRODUCED DRAWING
NO MANUAL ALTERATION

SCALE 1:10
INCHES
MILLIMETERS

THIS DRAWING IS PROPERTY OF AIRBUS DEFENCE AND SPACE S.A. WHICH RESERVES ALL RIGHTS			
SURFACE HOLDINGS NO. 000-0000		IDENTIFY EXT. ON DRAWING	
LIMITS NOT STATED	CLASSED	INTERCHANGEABLE PART	YES NO
CONTRACT		IDENTIFY INFL. PART	YES NO
DATE	10/01/2000	DESIGN	00000000000000000000
REV	00000000000000000000	REV	00000000000000000000
SCALE 1:10		TITLE	
DRAWING NO.		PROJECTION	
SHEET 01A		FROM A	

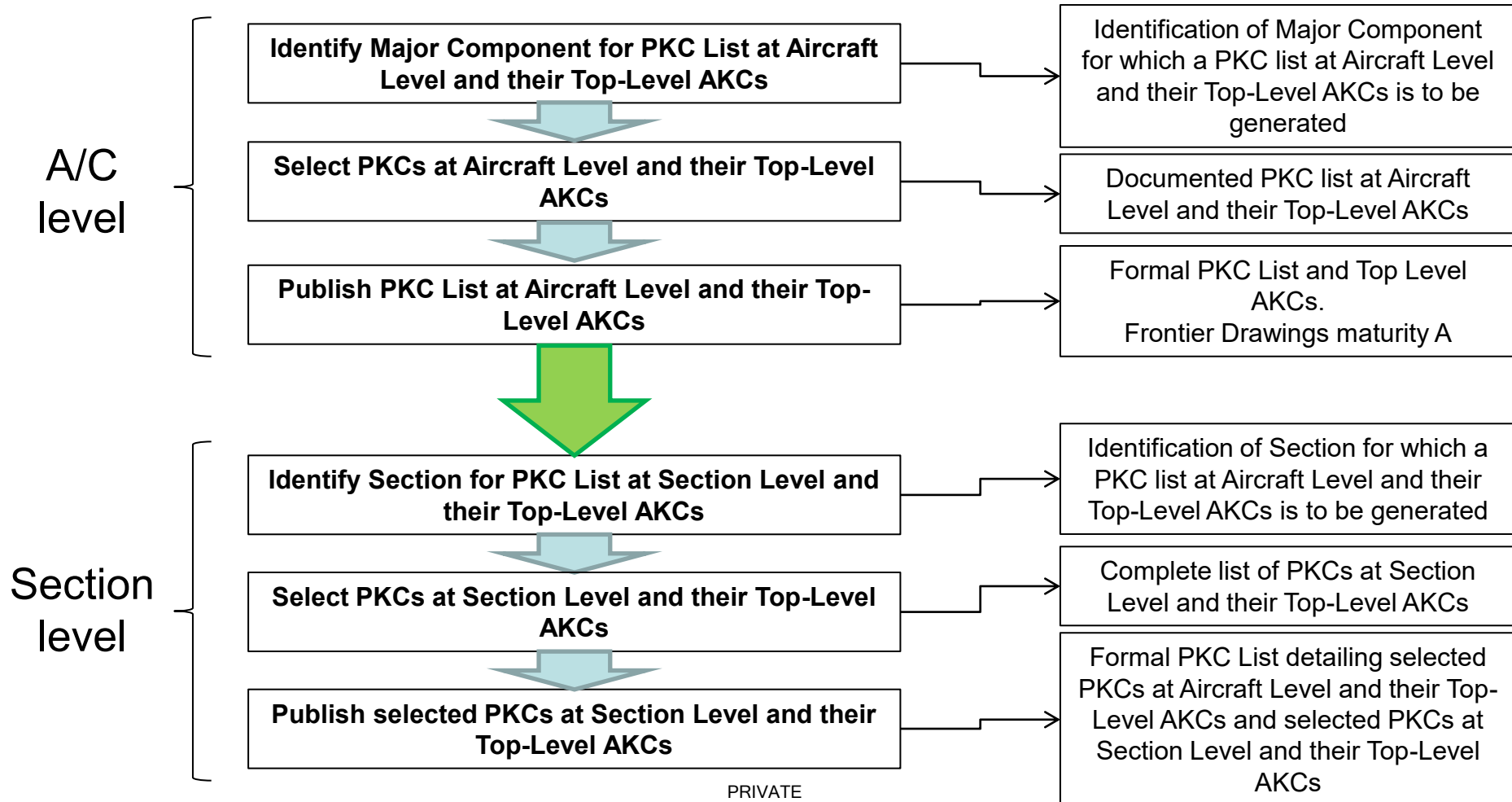
KC&T definition process

- The global A/C interface definition process is a top-down cascade from the PKC and top level AKCs to the individual MKCs of each part involved in each KC.
- All the design groups are involved in the definition of each tolerance taking into account the design principles, datum structure, assembly processes, tooling philosophies and manufacturing capabilities.
- Based on the data collected, the KC&T are evaluated using tolerance analysis for the validation and refinement of the values achieved.
- The KC&T process starts in the concept phase with the identification of the PKCs and top level ACKs till the development phase reviewing the feed-back from the final assembly line and flight tests.

KC&T definition process

— Selection of PKCs and Top-Level AKC

Outputs

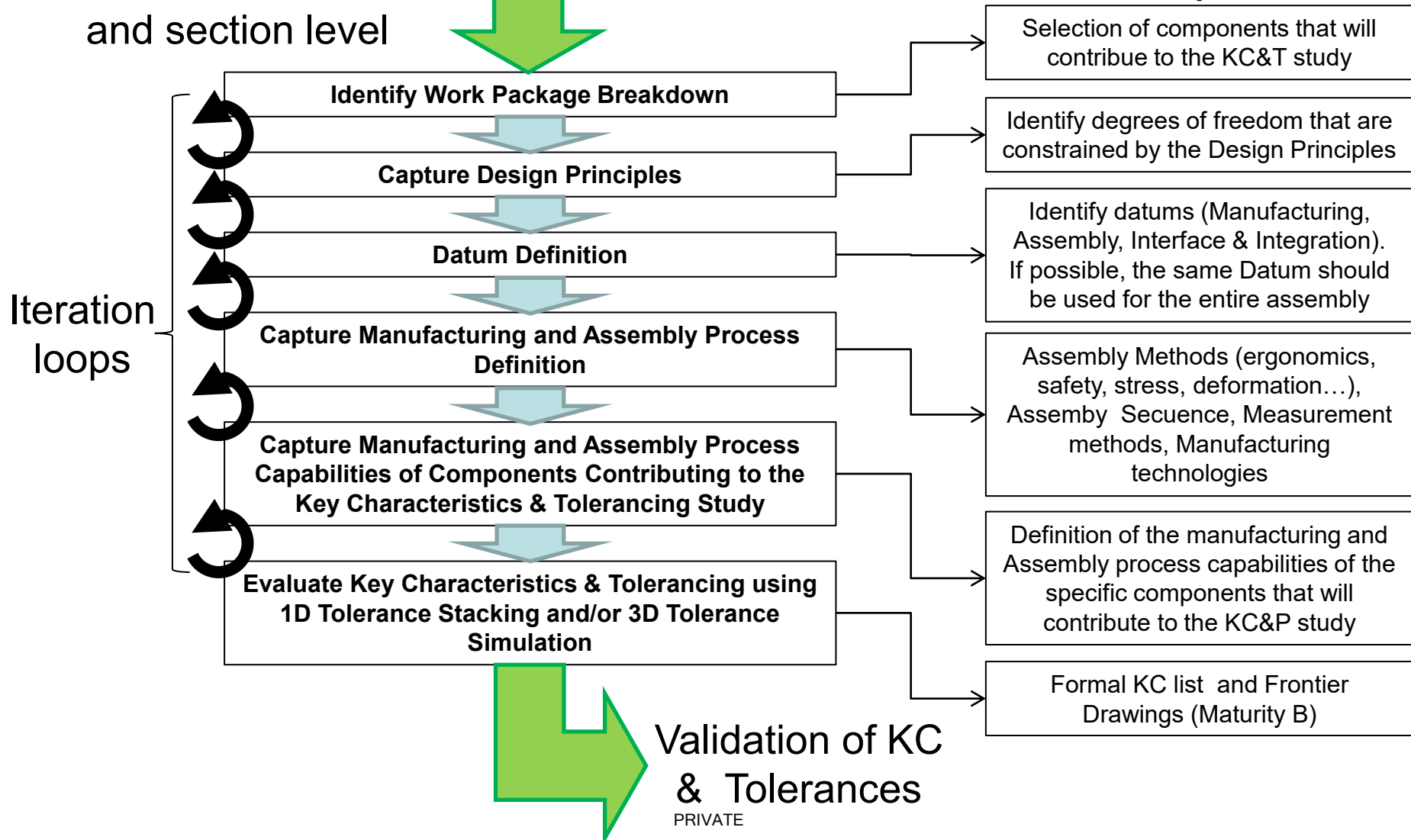


Interface Control Document

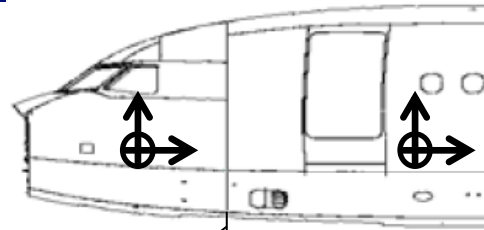
KC & Tolerances
selection at A/C
and section level

— Interface Product cascade

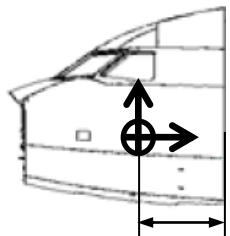
Outputs



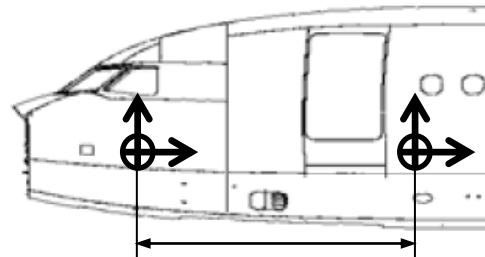
Interface Control Document



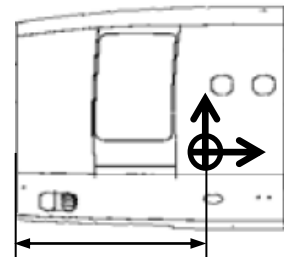
PKC $A \pm 2$



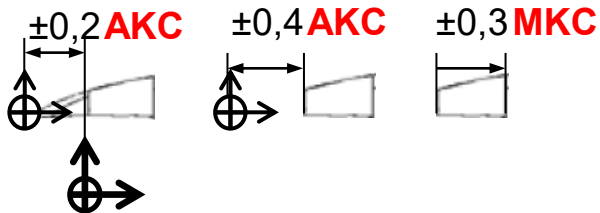
AKC $H \pm 0,9$



AKC $\pm 0,2$

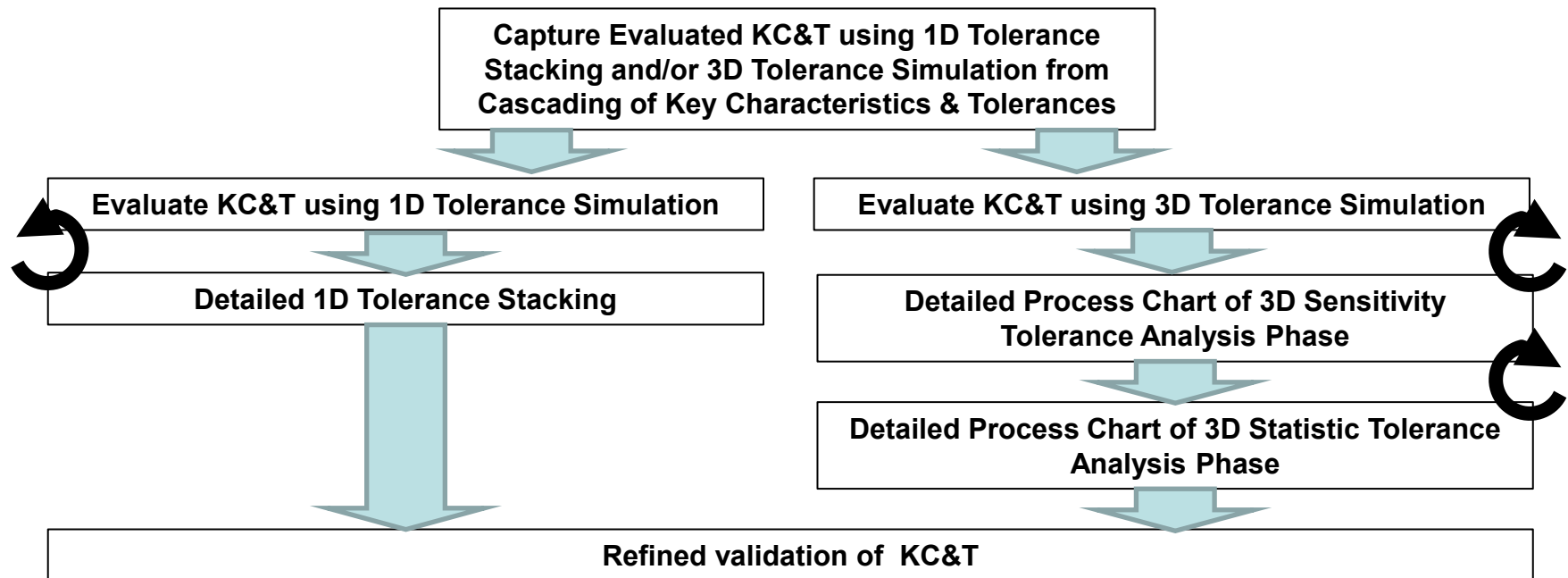


AKC $J \pm 0,9$



KC&T definition process

— Validation of KC&T



KC&T definition process

— Implementation of KC&T

- The implementation of Key Characteristics & Tolerances shall be commenced from the beginning of the Concept phase and continue until its completion at end of the Definition phase.
- The global A/C interface definition process is a top-down cascade from the PKC and top level AKCs to the individual MKCs of each part involved in each KC.
- The KC&T data embodied on the A/C definition drawings covers static build only, dynamic variations are not covered.
- The drawing set will only show KCs that are related to the functional features of the delivered part, assembly or tool.
 - The drawing set will not show all of the KCs within a cascade for a given PKC requirement.

KC&T definition process

— Implementation of KC&T

- There is a naming convention at Airbus level to facilitate the common understanding and rapid identification of the Key Characteristics & Tolerances applied onto the aircraft definition drawing set.
- It is not based on the ATA References. A "Geographical Reference" Section Numbers for the Major Components are used for this case.
- The identification of the KC in the documentation is done through a flag where:
 - **KC**: is the type of key characteristic under review (PKC, AKC, MKC or STR)
 - **xx** : Section number
 - **yy**: Section number (if it is within one section or MKC, it will be "00")
 - **nnnn**: KC number



KC-xx-yy-nnnn

Interface Control Document

— KC numbering

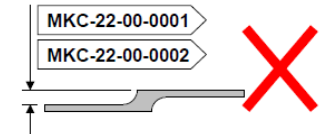
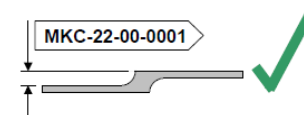
Major Component Name	Section Number	Section Name	Number
Fuselage	10	General Assembly	0001 - 9999
Nose Fuselage	11	Nose Cockpit Unit	0001 - 9999
	12	Upper Rear Unit	0001 - 9999
	1112	Lower Unit	0001 - 9999
	11-01	Nose Fairing	0001 - 9999
Forward Fuselage	13	Forward Forward Fuselage	0001 - 9999
	14	Rear Forward Fuselage	0001 - 9999
Centre Fuselage	15	Centre Fuselage	0001 - 7000
		Belly Fairing	7001 - 9999
	21	Centre Wing Box	0001 - 9999
Wing	20	General Assembly	0001 - 9999
	22	Outer Wing Box	0001 - 9999
	23	Leading Edge	0001 - 9999
	25	Trailing Edge	0001 - 9999
Wing Movable Parts	24	Slats	0001 - 9999
	26	Spoilers	0001 - 9999
	27	Flaps	0001 - 9999
	28	Ailerons	0001 - 9999
	29	Wingtips	0001 - 9999
Aft Fuselage	16	S16	0001 - 9999
	17	Forward Aft Fuselage	0001 - 9999
	18	Rear Aft Fuselage	0001 - 9999
Rear Fuselage	19	S19 & S19.1	0001 - 9999
VTP	31	VTP	0001 - 9999
HTP	35	HTP	0001 - 9999
Powerplant	40	Pylon/Nacelle General Assembly	0001 - 9999
	41	Pylon	0001 - 9999
	42	Nacelle	0001 - 9999
	45	Joint	0001 - 9999
Landing Gear	50	Undercarriage General Assembly	0001 - 9999
	51	Nose Landing Gear	0001 - 9999
	52	Main Landing Gear	0001 - 9999

KC&T definition process

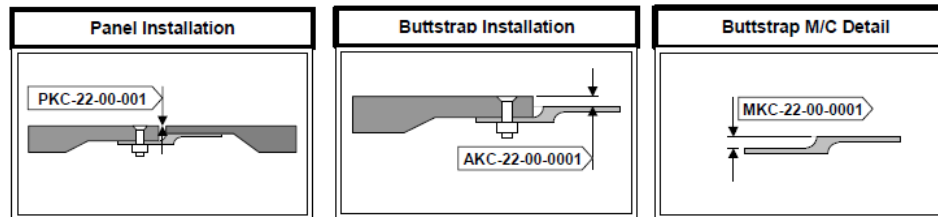
— Implementation of KC&T

▪ Rules

→ Only One Key Characteristic Flag per Feature



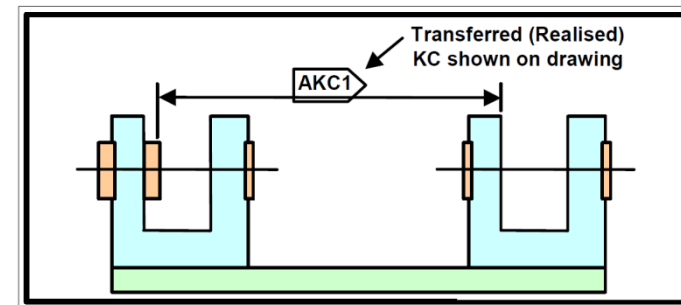
→ KC Only Shown on Drawing that Covers Point of Realisation or Deliverable Functional Features



→ Tolerance Shown for Every KC Flag

→ Transferred Tooling Features

- If a feature is achieved in the assembly by a tool, this KC is transferred to the drawing.

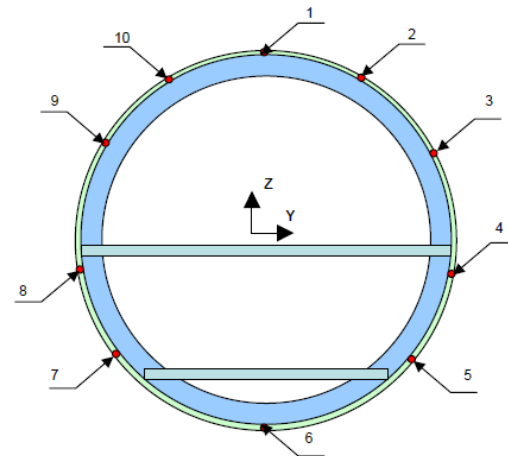


KC&T definition process

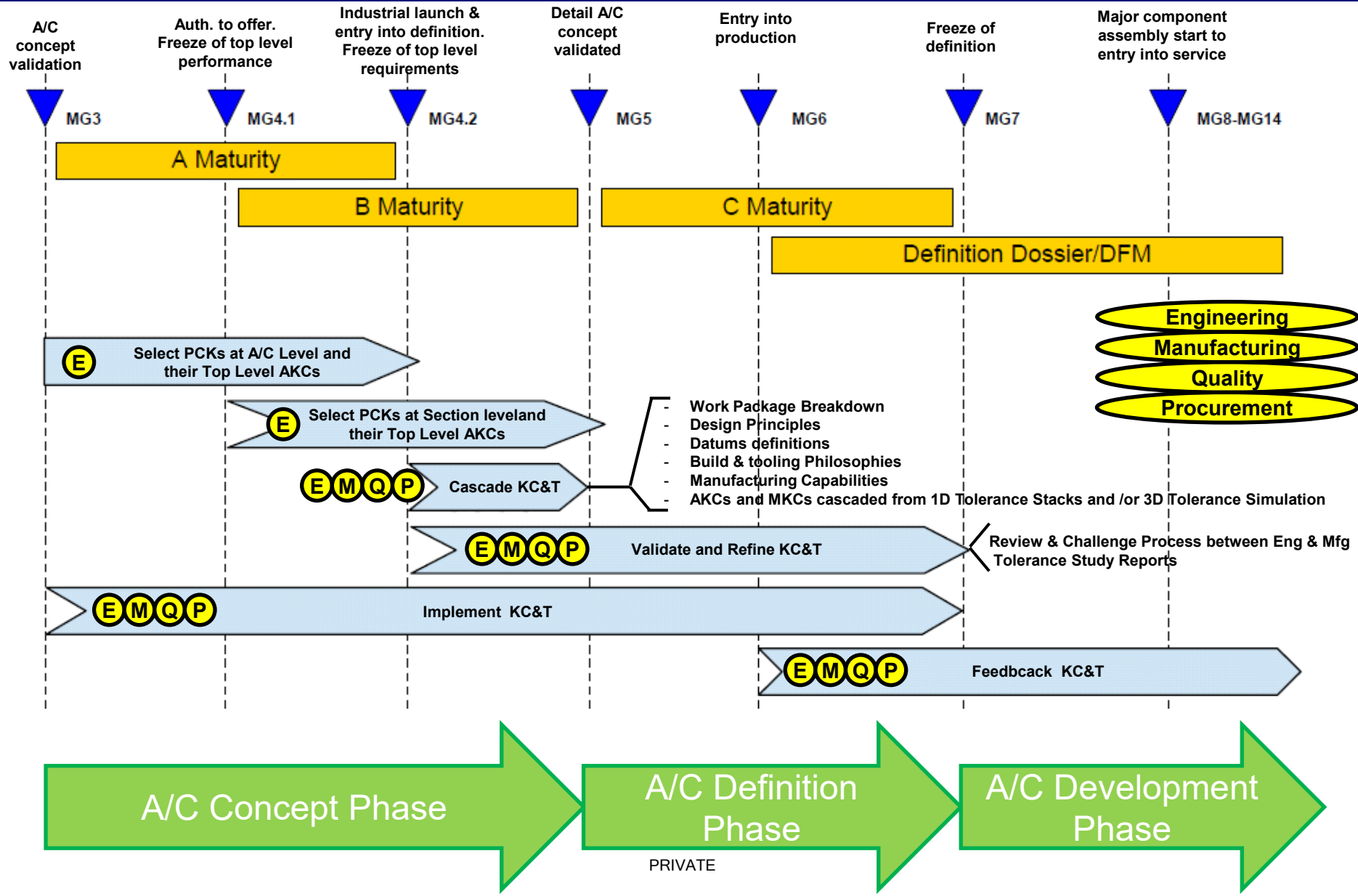
— Implementation of KC&T

- One KC can be along some complex or extended feature (plane, line, curve...), and some particular points shall be identified and be given their theoretical coordinates for each KC.
- The naming of such points shall be done by using the KC name plus the number of the point (i.e. for the AKC-15-21-0004 with 10 measurement points, you will have AKC-15-21-0004-01, AKC-15-21-0004-02, etc.)
- The tolerances described in the Frontier drawings shall be ensured and measured in the same configuration as during the integration of the component at the next assembly level above.

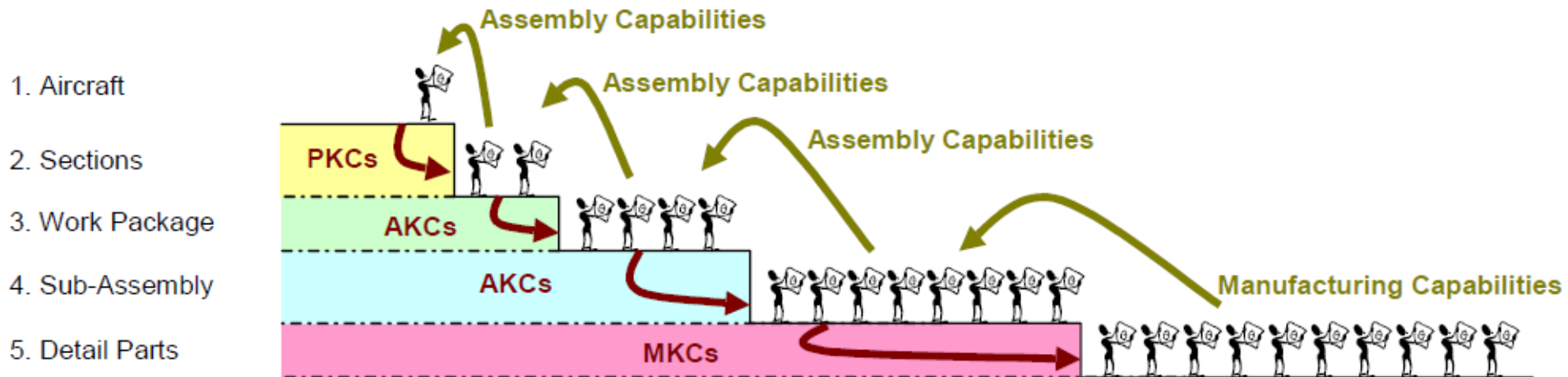
AKC-15-21-0004-01
AKC-15-21-0004-02
AKC-15-21-0004-03
AKC-15-21-0004-04
AKC-15-21-0004-05
AKC-15-21-0004-06
AKC-15-21-0004-07
AKC-15-21-0004-08
AKC-15-21-0004-09
AKC-15-21-0004-10



Interface Control Document



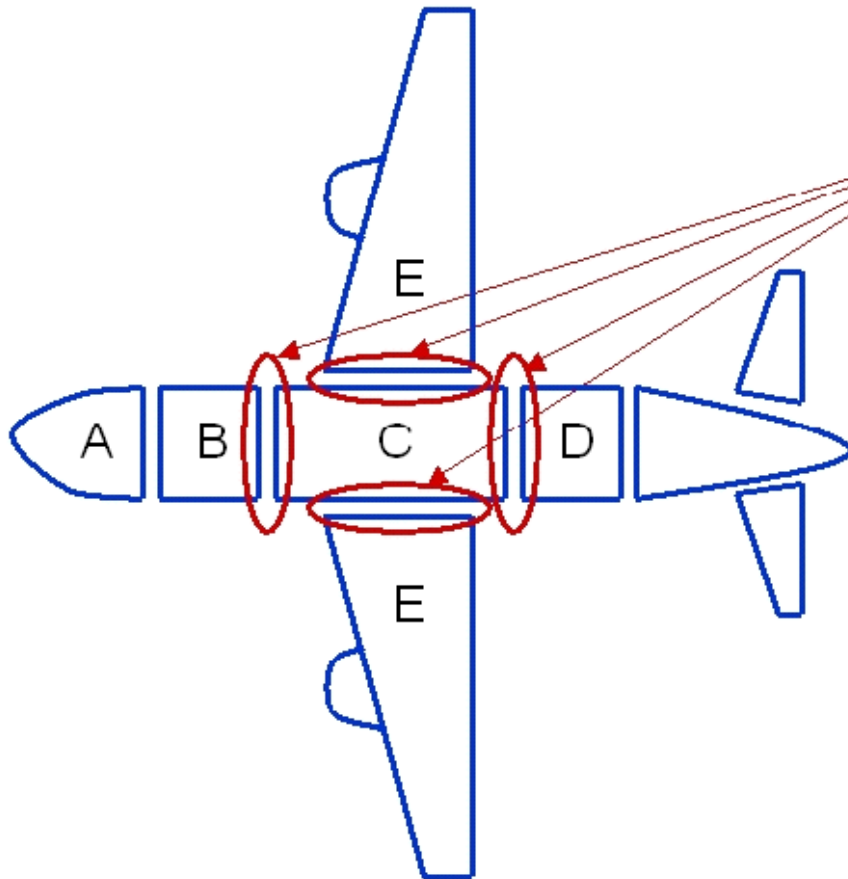
KC&T definition process



INTERFACE DOCUMENTS

Types of interface documents

- All the information related to the interfaces is share and agreed in different kind of documents and used to validate and fix information at a junction between A/C main sections and/or suppliers.
 - Frontier Drawing Matrix
 - It recapitulates all Frontiers and shows the breakdown of the aircraft following the assembly process.
 - Frontier Drawing
 - Defines the frontier between two or more Work Packages (WP) in terms of **tolerances** and requirements.
 - Package Frontier Drawing
 - Capture all tolerances for frontier boundaries of one WP. (Contractual)
 - Interface Drawing
 - The design principles that define the **geometry** between packages. (Contractual)
 - Junction Drawing
 - Define physical joint between packages.
- Prior to any data being embodied on the Interface documents, all functions within the Work package/Zone Team shall review and agree the Geometric Key Characteristics & Tolerancing and geometry information.



At the frontier, we need to study:

- Tolerances, physical datum
- Detailed description of the geometry and junction parts
- Holes position and properties, fasteners types
- Assembly and drilling allowances, CoS (Condition of supplier)
- Responsibilities/ownership
- ICY Contractual Interchangeability

Interface documents

— Frontier Matrix

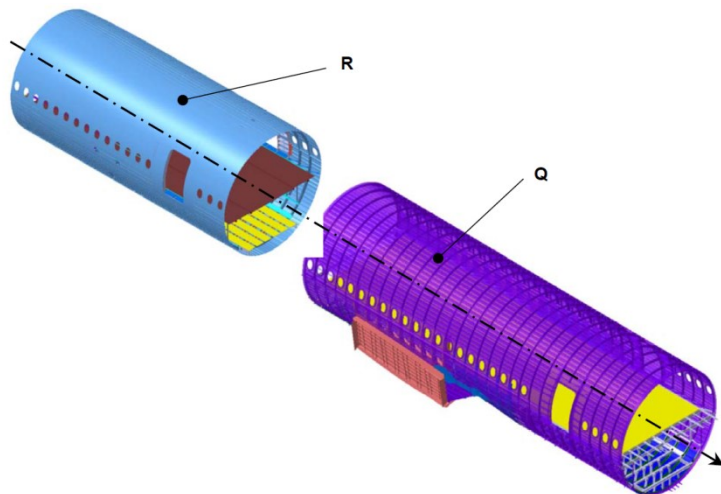
- Defines and indicates the complete list of Frontier Drawing and Package Frontier Drawing.
- Describes the complete macro assembly process.
- Specifies the responsibility of the establishment of Frontier Drawing and Package Frontier Drawing.
- Shows a general view of the area concerning the Frontier sharing.
- Specifies the manufacturing responsibility for each item.

frontier drawing responsible

package frontier drawing responsible

ITEM	ENGINEERING RESPONSIBILITY	FRONTIER DRAWINGS				CF	CF	RSP	CF	CF	CF	CF	CF	CF	CF	FF	FF	FF	FF	CF	PACKAGE FRONTIER DRAWINGS	package frontier drawing resp
		SUB-ASSY	ASSEMBLY 1	ASSEMBLY 2	Work Package (BAABI)	NZ										HAM				FAL		
A			Centre Wing Box		NT	X	X		X	X	X	X									VVVF21095	CF
B			Main Landing Gear Bay		RSP		X		X		X	X									VVVF21096	CF
C			Keel Beam + Aft Lower Shell		NT				X	X		X									VVVF21097	CF
			Rear Floor Grid																			
			Front Floor Grid																			
			Side panel																			
			Upper Panel																			
E			Upper Shell		RSP					X	X		X							O	VVVF21098	CF
F			Fwd Lower Shell		RSP						X	X								O	VVVF21099	CF
G			Lateral Panels		RSP															O	VVVF21100	CF
H			Belly Fairing		RSP																	
I			Doors		RSP																	
J			Pax Floors		RSP															O		
K			Cargo Floors		RSP															O		
Q				Section 15-21	CF															X		
L			Floor Grid		RSP											X	X			O	VVVF21101	FF
M			Lateral Shells		RSP												X	X		O	VVVF21102	FF
N			Lower Shell		RSP													X		O	VVVF21103	FF
O			Upper Shell		RSP															O	VVVF21104	FF
P			Doors		RSP																VVVF21105	FF
R				Section 13-14	FF																	
S				A/C	FAL															X		

Manufacturing responsibilities



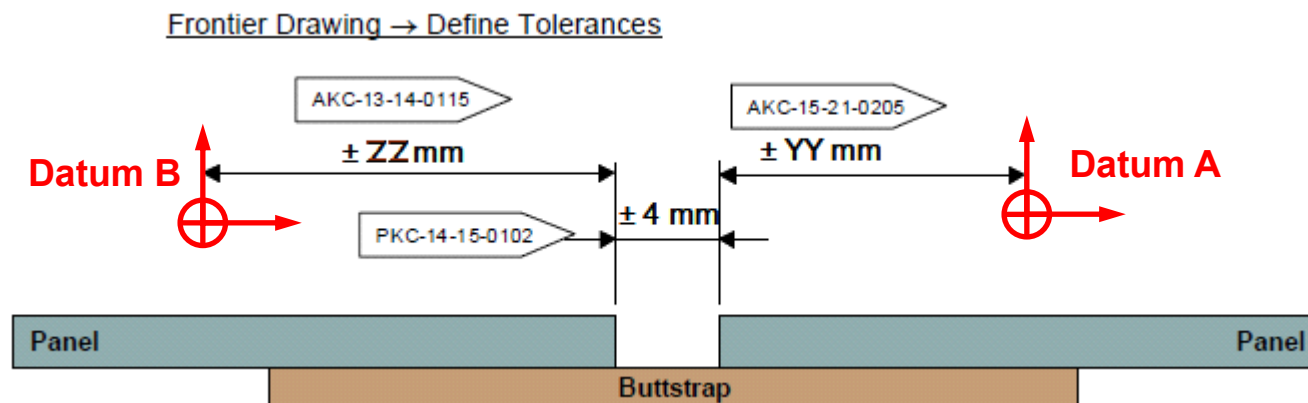
- The black case corresponds to a frontier drawing where a WP is added by one or more interfaces to another WP (see X and O in the same column)
 - The Frontier Drawing describes this or those interfaces
 - The Package Frontier Drawing summarises all the int
 X → The considered WP is in direct contact with the added WP
 O → A sub-element of the considered WP is in contact with the added WP (or one of the sub-element)

PRIVATE

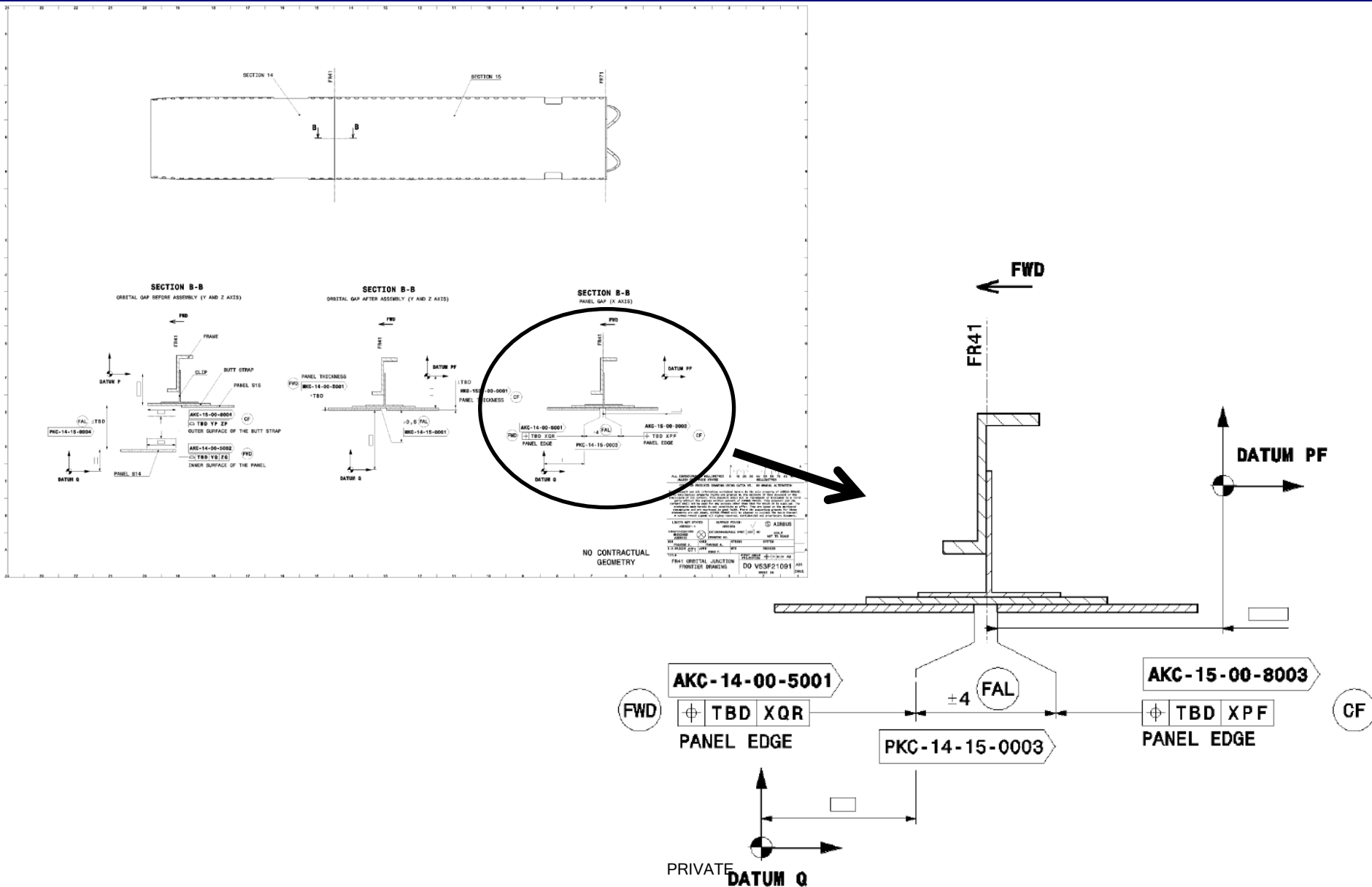
Interface documents

— Frontier Drawing

- It defines relationships between WP frontiers (two or more) in terms of tolerances and requirement, and it is linked to the assembly process.
- There is one frontier drawing per assembly stage (so one frontier drawing can contain several frontiers).
- Contains required overall tolerances and defines the share and ownership of tolerances between WPs.
- If necessary defines the physical datum of each package.



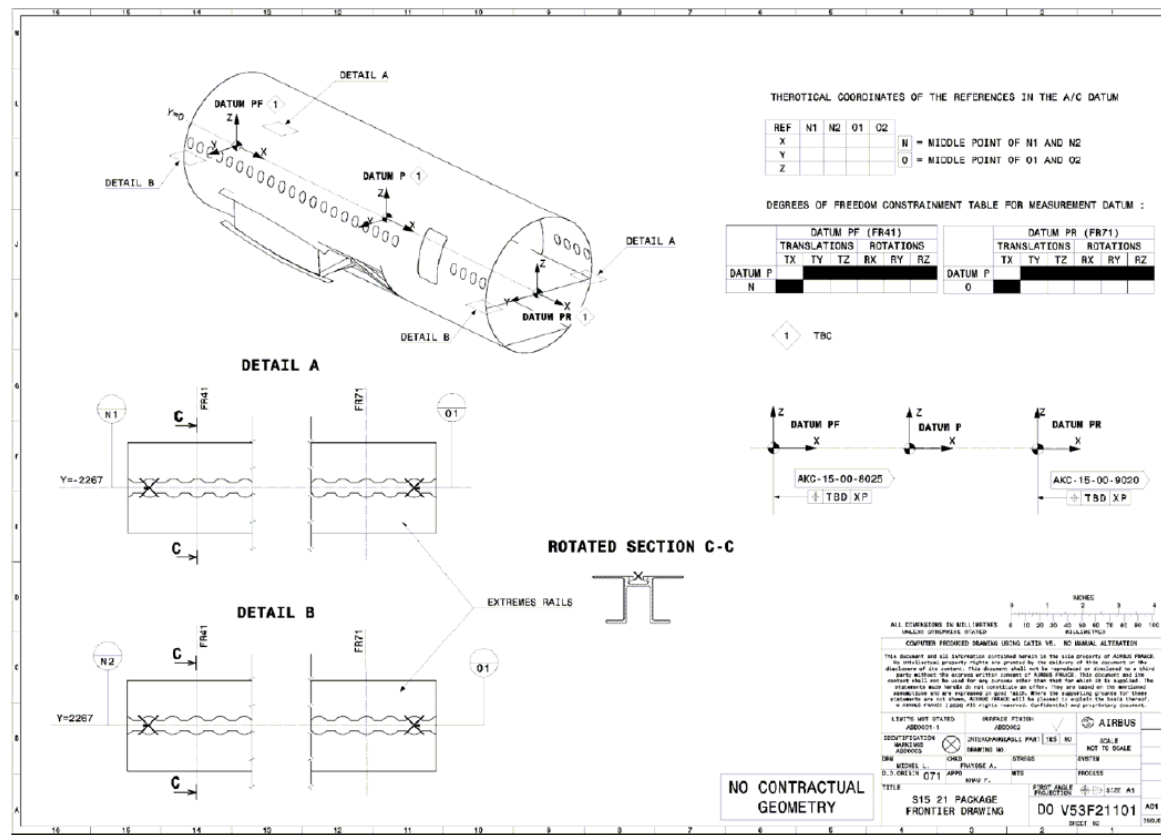
Interface Control Document

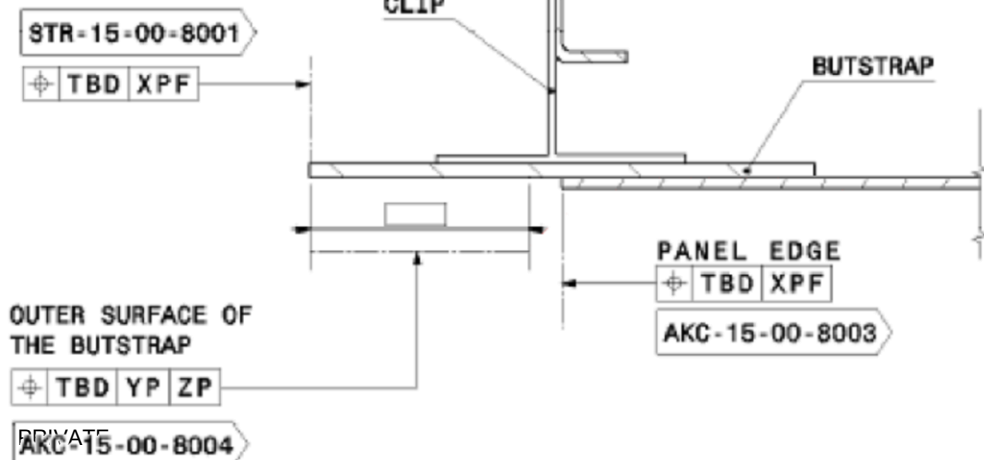


Interface documents

— Package Frontier Drawing

- Define the physical datum of one package.
- Captures tolerances at the frontier of the WP.

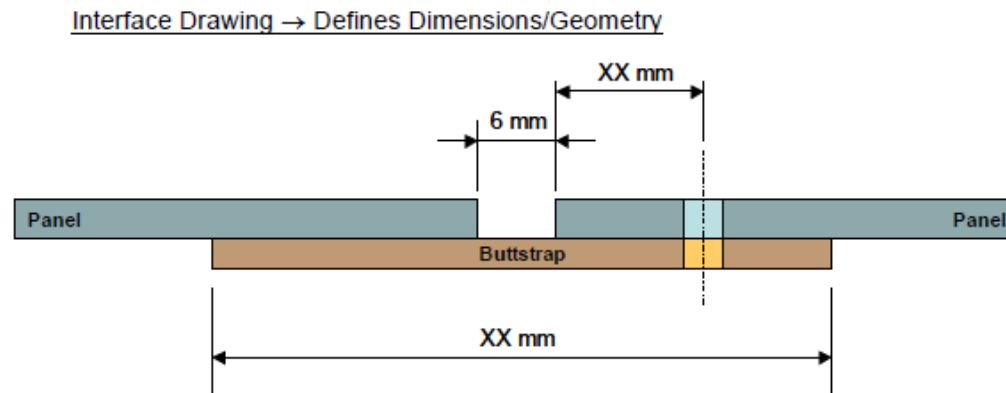




ICD concepts

— Interface Drawing

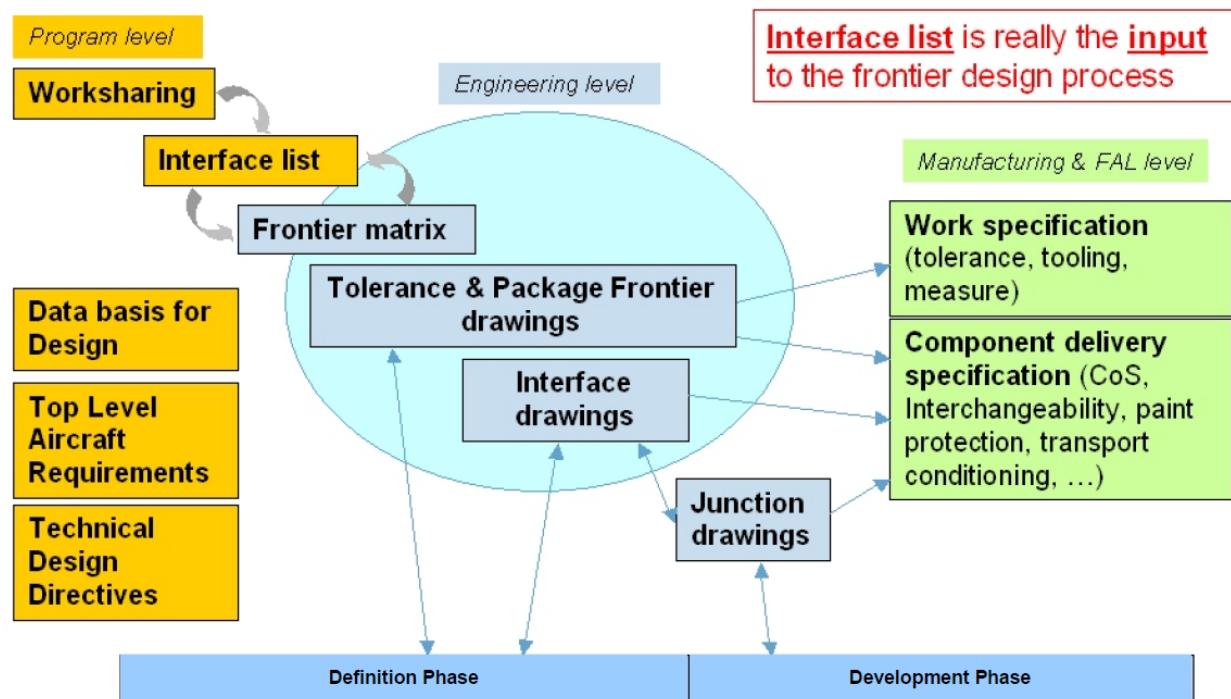
- Interface drawings define the interface contractual geometry between packages according to the design principles.
- It includes:
 - Detailed geometry definition (dimension on drawings or 3D data, tolerances),
 - Definition of holes position and fasteners types,
 - Definition of the space reserved for assembly and disassembly operations of spares (key overall dimensions, etc.),
 - Definition of torque-tightened fasteners, connecting rods fixed length (blocked) calculation, etc.

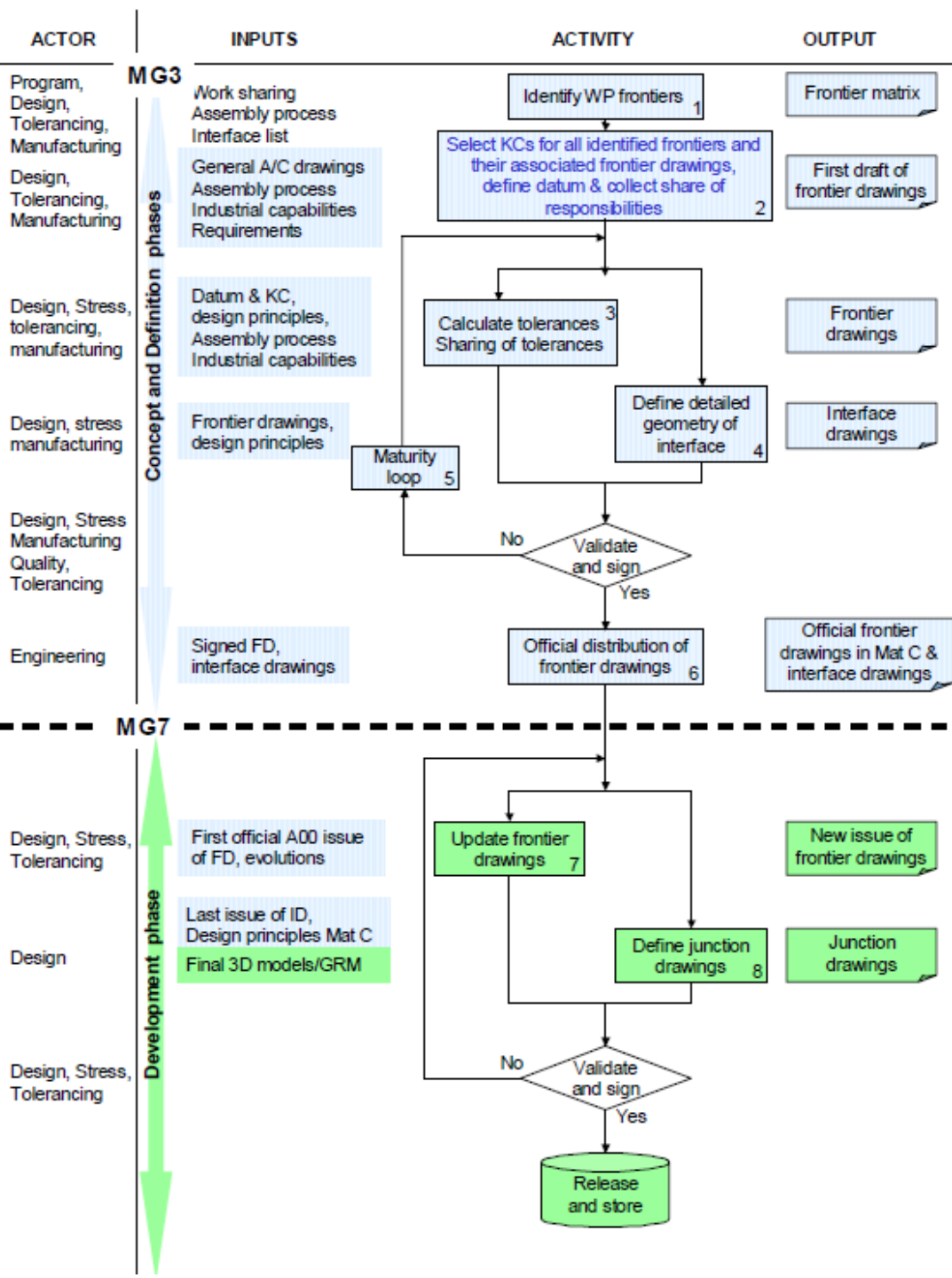


ICD concepts

— Junction Drawing

- Junction drawings define the assembly process between two WP.
- The information of the JD is extracted directly from the Interface drawings during the development phase when all the information is agreed and signed.





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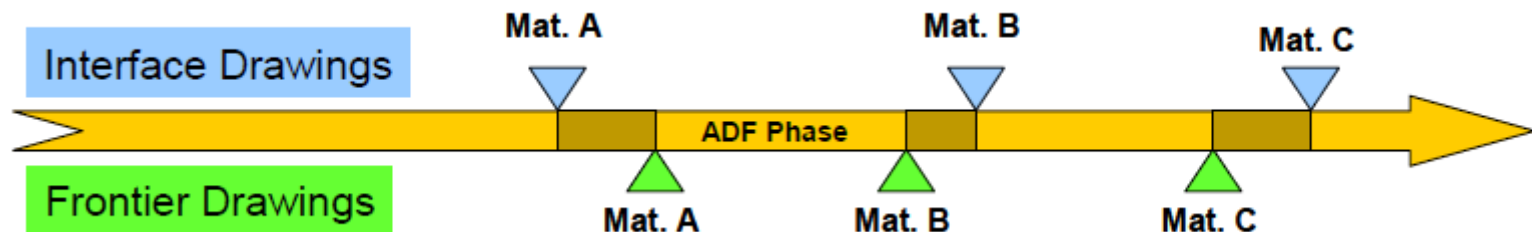
NO.	NAME OF ACTIVITY	DESCRIPTION
1	Identify WP frontiers	List all frontiers that have to be studied and the associated frontier drawings.
2	Select KCs for all identified frontiers and their associated frontier drawings, define datum & collect share of responsibilities	Select top-level KCs and STRs for all identified frontiers and their associated frontier drawings. Define as soon as possible physical datum.
3	Calculate tolerances Sharing of tolerances	Perform the tolerance analysis. From the results and the KCs and STRs, share the tolerances.
4	Define detailed geometry of interface	Detailed geometry study of the interface, as the normal design principle study.
5	Maturity loop	Modification of some information on FD and ID, depending on the need and the maturity of the study. We can also have an update of the inputs, particularly on WPs design principles.
6	Official distribution of frontier drawings	First release of frontier drawings (definition dossier drawings) and last issue of interface drawings (study drawings).
7	Update frontier drawings	In case of need, frontier drawings can be updated after MG7.
8	Define junction drawings	Junction drawings are done from interface drawings and GRM.

Geometric Key Characteristics & Tolerances Guaranty

- During the concept phase, all the teams and companies involved in the development of the elements shall demonstrate that tolerances will be ensured as mentioned in the Frontier Drawings.
- Data coming from Engineering Office as well as Manufacturing plants shall ensure that assembly process, jigs and tools, measurement means are in line with requirements embodied in the Frontier Drawings.
- For this purpose, tolerance stacks in 1D, in 3D, manufacturing capabilities through measurements, etc shall be the basis.

Interface management

- Interface and Frontier Drawings are managed in configuration by different maturities (A, B & C) according to A/C development till definition Phase where are frozen.
- Relationship between Interface Drawing and Frontier Drawing
 - For maturity A, Frontier Drawings need rough design principle in order to initiate KC Lists (identification of WPs, requirements between WPs, draft datum, etc.) and tolerance study. Thus, Interface Drawing Mat A shall be released before Frontier Drawing Mat A.
 - Afterwards, Maturity B and C of Interface Drawings shall take into account tolerance analysis in the Design Principles and thus take into account Frontier Drawings. As a matter of fact, Frontier Drawings Maturity B shall be released before Interface Drawings Mat B and the same goes for maturity C.



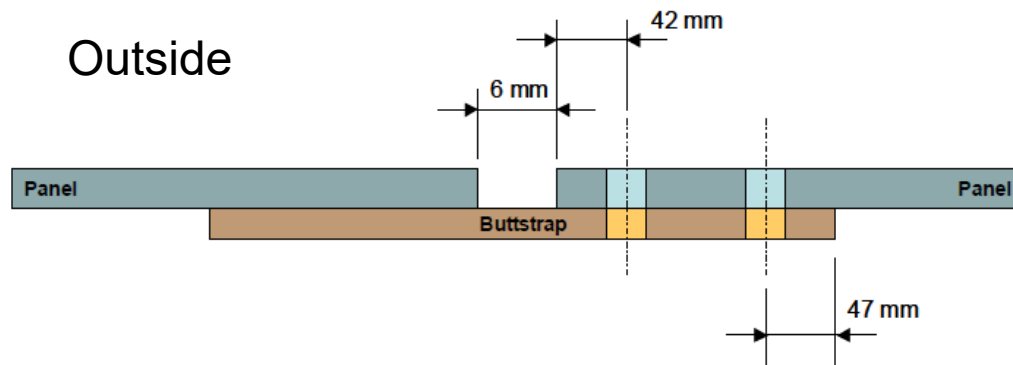
ITEM	FRONTIER MATRIX	FRONTIER DRAWING & PACKAGE FRONTIER DRAWING	INTERFACE DRAWING
Maturity A Deliverables	- Shows a general view of the area concerning the Frontier sharing - WP item work sharing - Definition of repere items - Macro Assembly process - Concept Phase Drawing numbering - Drawing signatories identified	- Global WP breakdown; datums - Frozen PKC requirements (no value mandatory at this stage) - Assembly process and associated tooling & measurement means identified - Drawing signatories identified - Signature <u>non mandatory</u>	Define parts ownership, rough geometry of the junction.
Maturity B Deliverables	- Identification of the Drawing realization responsibility - Validation of macro assembly process	- Frozen datums, Work sharing, assembly process - Tolerance stack chaining started - First stress check - Signatures <u>mandatory</u>	- Detail the interface geometry. - Specify types, position and ownership of H&F. - Specify CoS.
Maturity C Deliverables	- Definition Phase Drawing numbering - All signatures collected	- Calculated and agreed tolerances - AKC and MKC values - Frozen tolerance stack chaining - Measurement means frozen in accordance with tolerance values in Frontier Drawings - Signatures <u>mandatory</u>	- Finalize detailed geometry of the interface. - Finalize H&F definition: type, position, and ownership. - Finalize CoS definition. - Stress finish the check of the interface.

Relationship between Interface Drawing and Frontier Drawing

The design principle and CoS selected during the concept phase have an impact in the final definition of the junction.

Example:

Orbital Junction between two sections. Holes for fasteners are roughly positioned in the first figure.



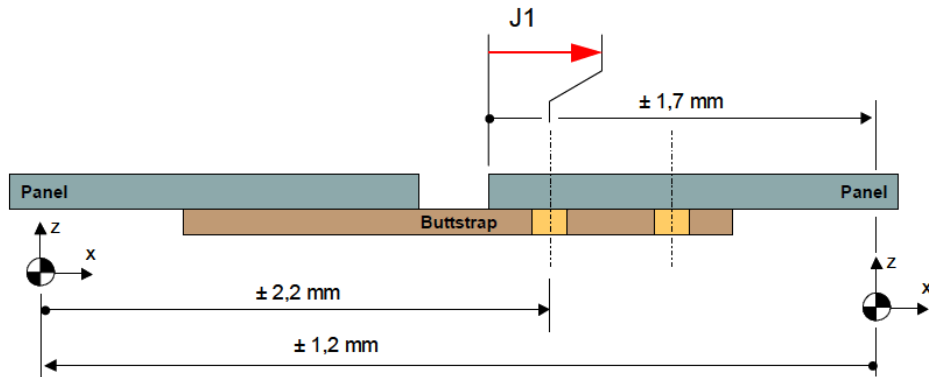
Then two assembly process is possible:

- Concept 1: hole already drilled in buttstrap, then counter drilling from inside to outside
- Concept 2: hole already drilled in panel then counter drilled from outside to inside

In each scenario, Tolerance analyses have an impact on the final geometry.

Relationship between Interface Drawing and Frontier Drawing

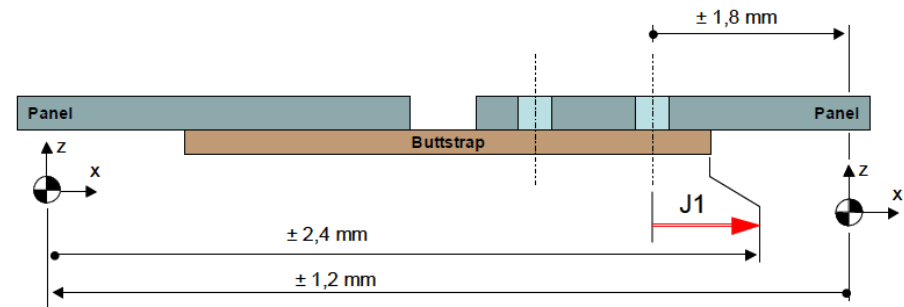
Concept 1



Tolerance analysis:

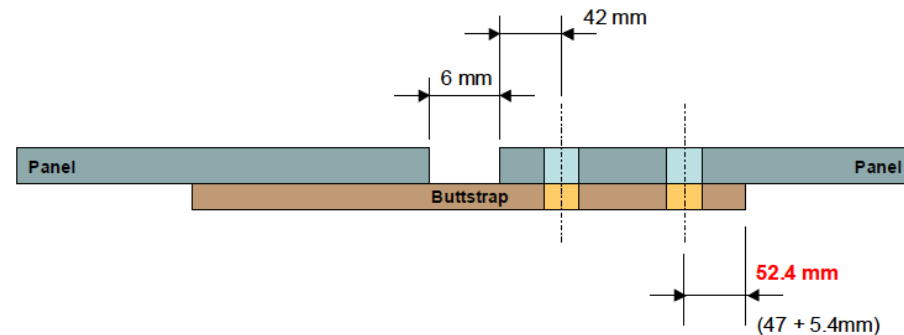
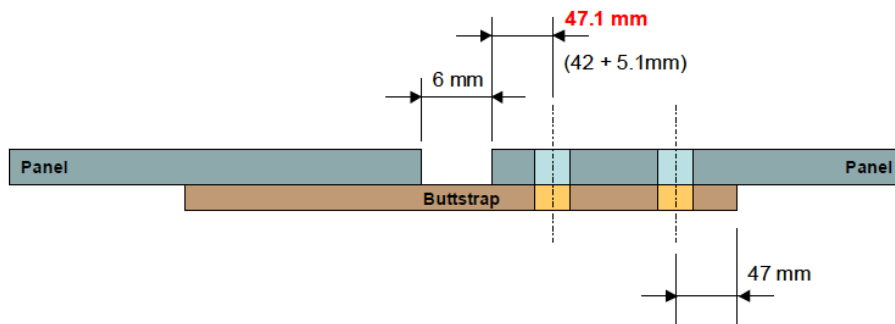
$$J1 = \pm 2,2 \pm 1,2 \pm 1,7 = \pm 5,1$$

Concept 2



Tolerance analysis:

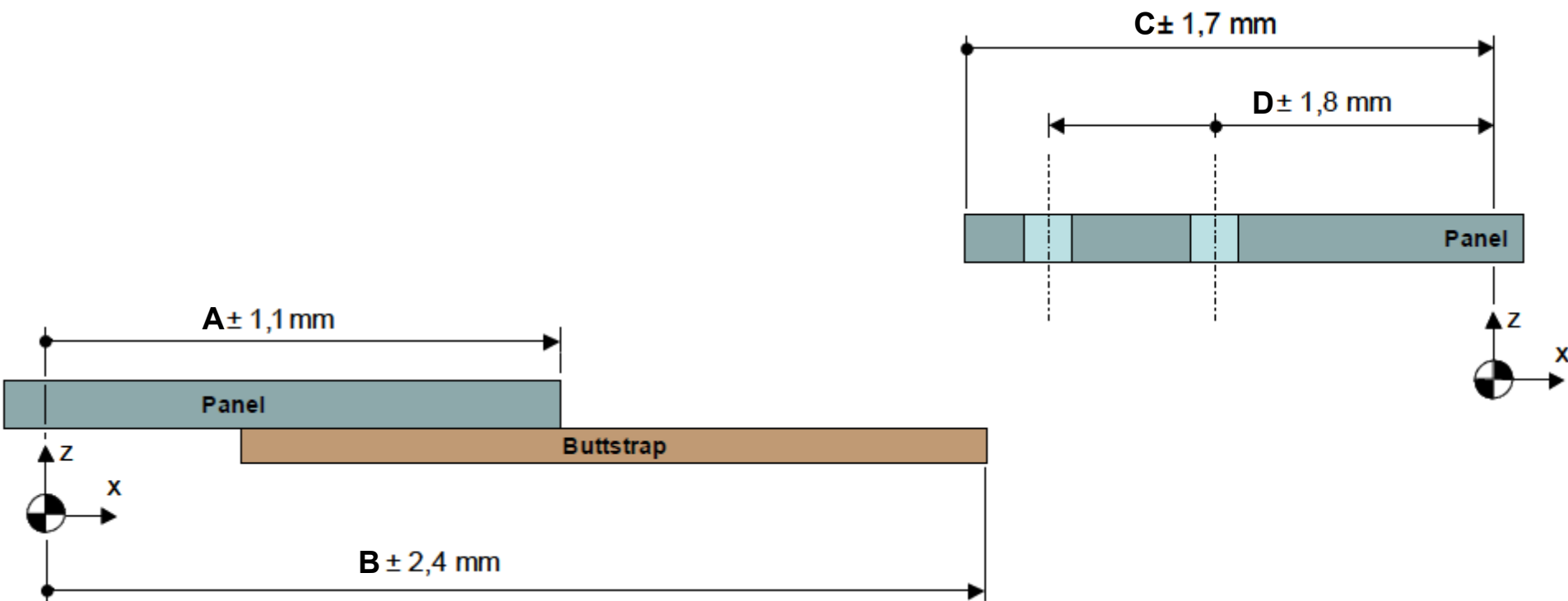
$$J1 = \pm 2,4 \pm 1,2 \pm 1,8 = \pm 5,4$$

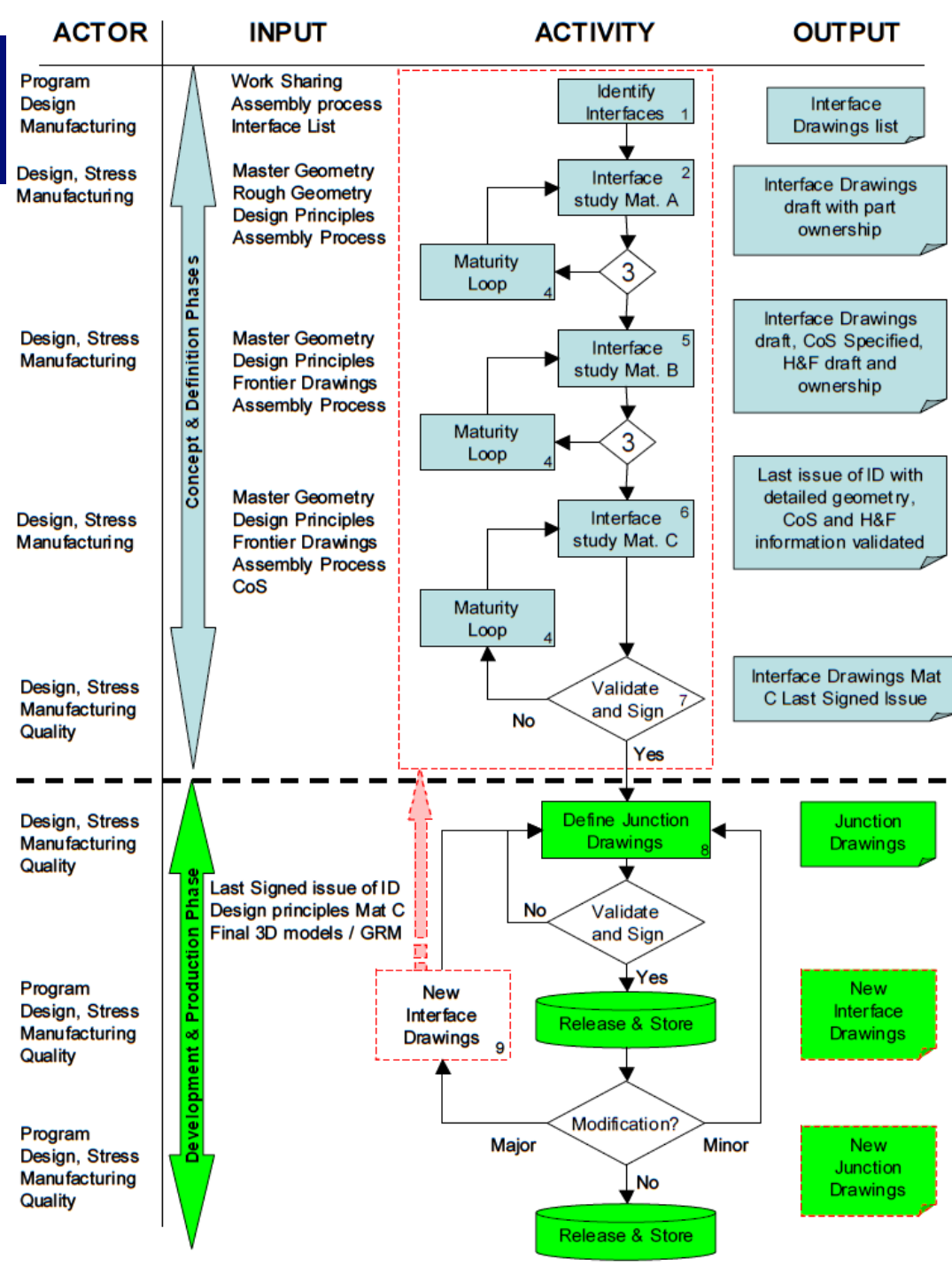


Relationship between Interface Drawing and Frontier Drawing

The information included in the Interface Drawings are provided by each actor in the interface.

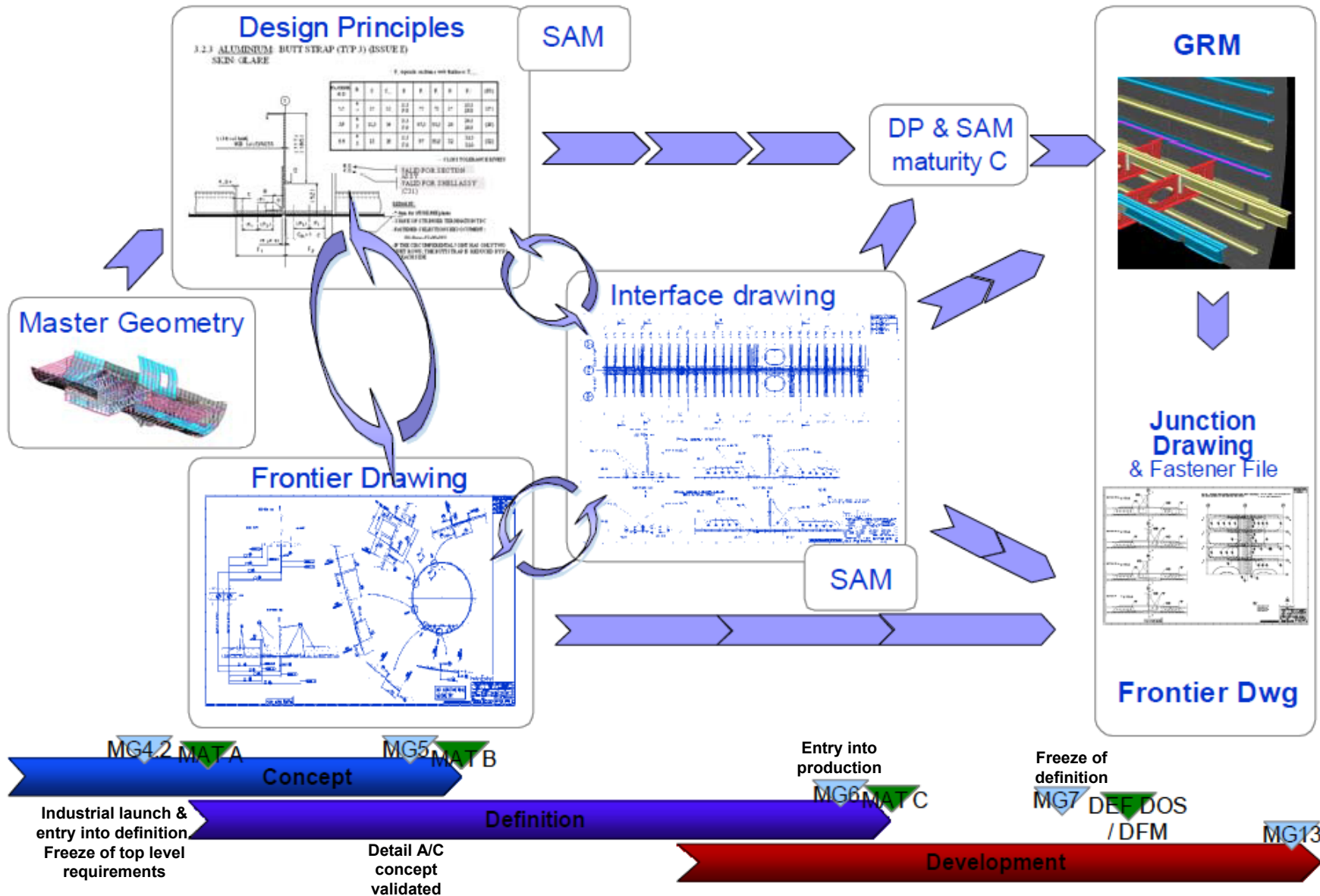
Each actor must provide the information of their own parts or assemblies. All this data is included in a document called Junction Drawings.





No.	Name of Activity	Description
1	Identify interfaces	List all interfaces that have to be studied and the associated interface drawings (work breakdown between teams and designers)
2	Interface Study maturity A	Define parts Owner ship, rough geometry of the junction
3	Intermediate validation of the maturity	Validate the frame of work corresponding to the current maturity: agreement between interfacing parties (engineering and manufacturing) on the current design. The target is to allow packages to work inside defined borders
4	Maturity Loop	Modification of some information on interface drawings depending on the need and the real maturity or the study. We can also have an update of the inputs of the interface study, particularly on the interfacing packages design principles.
5	Interface study maturity B	Detail the interface geometry. Specify types, position and ownership of H&F Specify CoS
6	Interface study maturity C	Finalize detailed geometry of the interface Finalize H&F definition: type, position, ownership Finalize CoS definition Stress finish the check of the interface
7	Validate and sign	End of the study: all parties agree on the whole design of the interface
8	Define junction drawings	Junction drawings are done based on the interface drawings and GRM inputs
9	New Interface Drawings	In case of major modification, new Interface Drawings are needed then new cycle (1 to 8) for creation of new Interface and Junction Drawings

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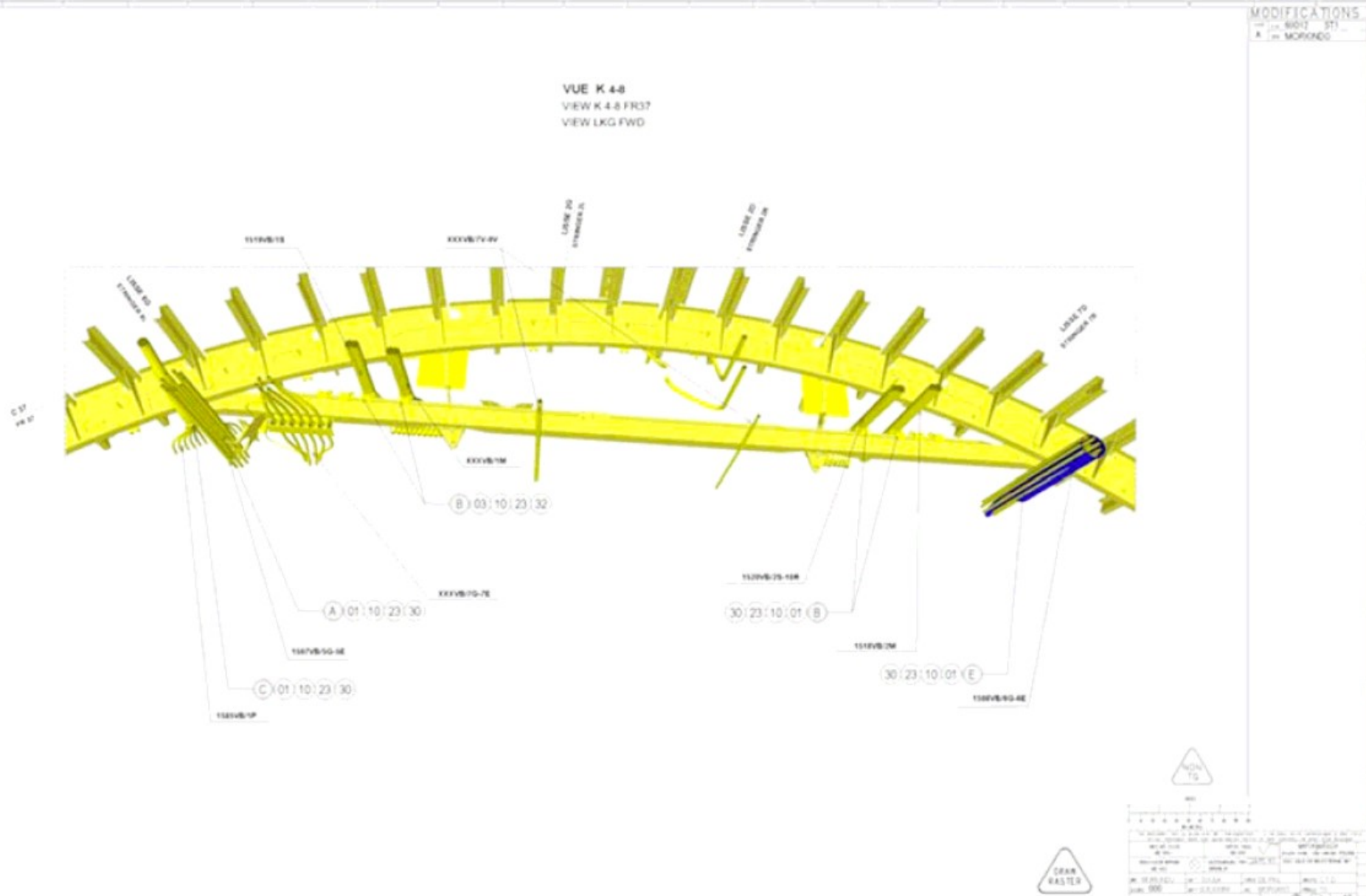
Frontier drawings for non structural elements

- Systems can be divided in two main groups:
 - "Structural systems": can be structure ATA or other ATA
 - Ailerons/spoilers actuators,
 - Landing gears, ...
 - Case 2: "Non structural Systems": systems routing across the aircraft
 - Tubing: hydraulics, pneumatics, fuel, oxygen...
 - Electrical,
 - Air conditioning, etc.
- For "Structural systems", the Structure Frontier Design process can be applied as define in the sheets before as they use to be located in one specific area.
- For "Non structural systems", as they are installed across several WPs with a lot of interfaces, inside the system itself and also with structure from different WPs, a dedicated process is defined.

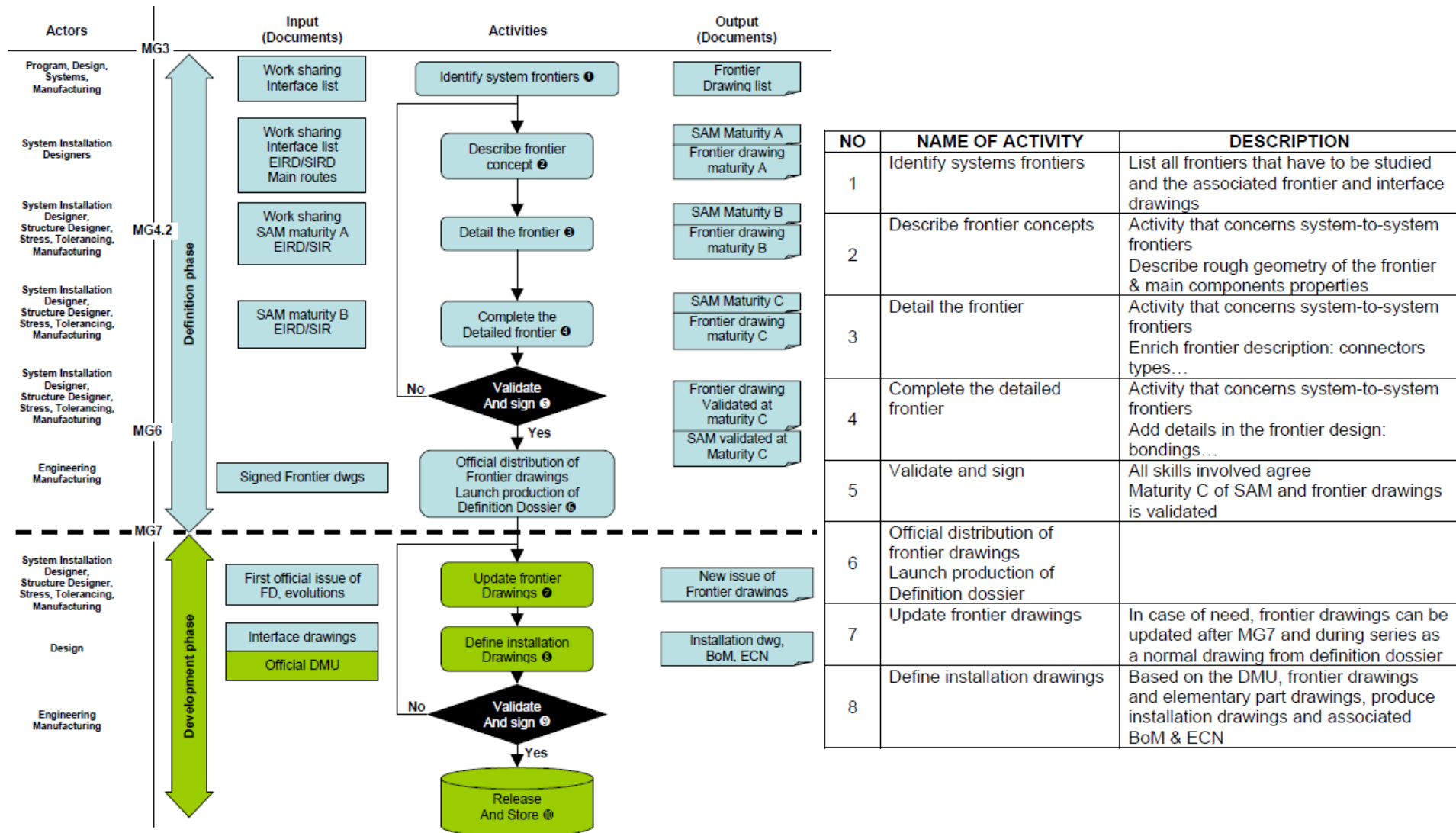
Frontier drawings for non structural elements

- The aim of this drawing is to define the frontier in term of:
 - Detailed worksharing between the CoEs and global sub-contractors: for each item it defines who produces the drawing, who supplies the item, who produces the assembly drawing, who does the installation,
 - Interface points coordinates and the reference to which they are given,
 - Connectors localisation and types, locating pin & master keys requirements,
 - Bonding (tag leads, ...),
 - "keep out" zones requests when required,
 - Conditions of supply when required (overlength, protection, ...),
 - Tolerances sharing when necessary.
- The geometry inside the system installation frontier drawing is non contractual so Picture Drawings can be used.
- There can be also technical notes associated to the System Installation Frontier drawings to justify tolerance calculation.

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Frontier drawings for non structural elements - maturities

- In terms of maturities, the main information needed in each step is as follows:
- Maturity A:
 - Describe frontier concept: where systems are broken, rough geometry of the frontier, main components properties, principles of installation
 - Global worksharing
 - Frozen requirements (no value mandatory at this stage), datums
 - Drawing signatories identified
- Maturity B:
 - Enrich the frontier description: connectors localisation & types, identification of bonding needs...
 - Frozen physical datums, assembly process and interface points coordinates
 - Tolerance stack chaining started (if needed)
 - Detailed worksharing (fittings, mobile brackets, etc)
 - "Keep out" zones request, Conditions of Supply (overlength, protection, ...) when needed
 - First stress check

Frontier drawings for non structural elements - maturities

— Maturity C:

- Complete the detailed frontier/add details in the frontier design:
 - Validated Bondings (tag leads), locating (locating pin, master key, ...), Interfacing points coordinates
 - Connectors defined with protection & overlenght (CoS)
 - Keep out zone validated where required
 - Validated detailed Worksharing
- Calculated and agreed tolerances (if needed) & physical datums,

1. Airbus internal documentation
2. ISO1101